Homework 29: 10.13-10.15 **Name** _____

AP Calculus BC **Due** _____

Period _____ **Score** _____ **/ 100**

Skill Builder 10.13 – Radius and Interval of Convergence of a Power Series

For the following, find the interval of convergence of the power series. Check for convergence at the endpoints.

1.) $\sum_{n=0}^{\infty}\left(\frac{x}{2}\right)^n$	**2.)** $\sum_{n=1}^{\infty}\frac{(-1)^n x^n}{n}$
3.) $\sum_{n=0}^{\infty}\frac{(2x)^n}{n!}$	**4.)** $\sum_{n=0}^{\infty}\frac{(3x)^n}{(2n)!}$

5.) $\sum_{n=0}^{\infty}(2x)^n$	**6.)** $\sum_{n=0}^{\infty}(-1)^n \frac{x^n}{n+1}$
7.) $\sum_{n=1}^{\infty}\frac{(2x)^n}{n^2}$	**8.)** $\sum_{n=0}^{\infty}\frac{(2x)!}{n!}$

9.) Determine the values for which the series $\sum_{n=1}^{\infty}(-1)^n(n!)(x-3)^n$ converges.	**10.)** Determine the radius of convergence for $\sum_{n=1}^{\infty}\frac{n^n x^n}{n!}$
11.) If the radius of convergence of the power series $\sum_{n=0}^{\infty}a_n\left(x-3\right)^n$ is $R = 5$, which of the following statements are ALWAYS TRUE, POSSIBLY TRUE, or ALWAYS FALSE. (a) The series converges for $x = 2$. (b) The series converges for $x = 7$. (c) The series converges for $x = 8$. (d) The series converges for $x = -6$. (e) The series converges for $x = -2$.	**12.)** If $\sum_{n=0}^{\infty}a_n\left(x-2\right)^n$ converges for $x = 5$, at what other numbers x must the series converge?

Topic 10.14 – Finding Taylor or Maclaurin Series for a Function

For the following, use the definition to find the Taylor (or Maclaurin) series centered at c for the function. When writing your answers, be sure to list the first 4 non-zero terms and the general term.

1.) $f(x)=e^{3x},\ c=0$	**2.)** $f(x)=e^{-3x},\ c=0$
3.) $f(x)=\sin 2x,\ c=0$	**4.)** $f(x)=\tan x,\ c=0$ (first two nonzero terms)

5.) $f(x) = \sin x,\ c = \frac{\pi}{4}$	**6.)** $f(x) = \cos x,\ c = \frac{\pi}{4}$

Skill Builder 10.15 – Representing Functions as Power Series

Write a power series for the function, centered at *c*.

1.) $f(x)=\dfrac{3}{x+4},\ c=0$	**2.)** $f(x)=\dfrac{5}{2x-1},\ c=0$
3.) $f(x)=\dfrac{-1}{x-3},\ c=6$	**4.)** $f(x)=\dfrac{3}{4-x},\ c=-1$
5.) $f(x)=\dfrac{3x}{x^2+x-2},\ c=0$	**6.)** $f(x)=\dfrac{4}{4+x^2},\ c=0$

Find the Maclaurin series for the following functions by using the table of power series from your notes or text. When writing your answers, be sure to list the first four non-zero terms and the general term.

7.) $f(x) = e^{-x^2}$	**8.)** $f(x) = e^{-4x}$
9.) $f(x) = \sin 3x$	**10.)** $f(x) = x\cos x$
11.) $f(x) = \cos x^{3/2}$	**12.)** $f(x) = \dfrac{\sin x}{x}$